Two country: h / f

- 参数相互独立
- 设定对称
- 冲击相互影响

Firm Setting

$$y_t = \left(m_t^{-v} + \sigma \cdot z_t^{-v}\right)^{\frac{-1}{v}}$$

$$\begin{cases} m_t = A_t \cdot k_t^{\theta} \cdot n_t^{1-\theta} & \text{Value added} \\ z_t : \text{inventory} \end{cases}$$

$$\Rightarrow \begin{cases} MPZ_t = \underbrace{\left(m_t^{-v} + \sigma \cdot z_t^{-v}\right)^{\frac{-1}{v}-1}}_{y_t^{v+1}} \cdot \sigma \cdot z_t^{-v-1} \\ MPK_t = \left(m_t^{-v} + \sigma \cdot z_t^{-v}\right)^{\frac{-1}{v}-1} \cdot m_t^{-v} \cdot \theta \cdot \frac{1}{k_t} \\ MPL_t = \left(m_t^{-v} + \sigma \cdot z_t^{-v}\right)^{\frac{-1}{v}-1} \cdot m_t^{-v} \cdot (1-\theta) \cdot \frac{1}{n_t} \end{cases}$$

HH Setting

$$E_0 \cdot \sum_{t=0}^{\infty} \beta^t \cdot U(c_t, h_t)$$

$$U = \frac{(c_t^{\mu} \cdot h_t^{1-\mu})^{\gamma}}{\gamma} \qquad \begin{cases} 0<\mu<1 \\ \gamma<1 \end{cases}$$

$$h_t = \sum_i \alpha_i \cdot l_{t-i}$$ （过去的闲暇会影响效用：加权平均）

$$= \sum_i \alpha_i \cdot (1 - n_{t-i})$$

权重 α_i: $\begin{cases} \sum_i \alpha_i = 1 \\ \alpha_i = (1-\delta)^{i-1} \cdot \alpha_1 \quad \text{衰减} \end{cases}$

$$= \sum_i \alpha_i - \sum_i \alpha_i n_{t-i}$$

$$= 1 - \alpha_0 \cdot n_t - \sum_{i=1} \alpha_i \cdot n_{t-i}$$

$$= 1 - \alpha_0 n_t - \alpha_1 \cdot \sum_i (1-\delta)^{i-1} \cdot n_{t-i}$$

$$= 1 - \alpha_0 n_t - (1-\alpha_0) \cdot \delta \cdot \boxed{\sum_i (1-\delta)^{i-1} \cdot n_{t-i}}$$ 含 a_t

简化为 α

$$\Rightarrow a_{t+1} = (1-\delta) a_t + n_t$$

$$= 1 - \alpha n_t - (1-\alpha) \delta \cdot a_t$$

代表过去的劳动

Capital build

资本建造需J期

分期付款：ϕ_j

$$k_{t+1} = (1-\delta)k_t + S_{1t}$$

$$S_{j,t+1} = S_{j+1,t}$$ （t期剩j+1期完工的项目）

$$x_t^i = \sum_j^J \phi_j \cdot S_{jt}^i$$ （t期所有在建项目付款和）

Tech shock

$$\lambda_{t+1} = A \cdot \lambda_t + \epsilon_{t+1}$$ （允许相互外溢）

$$\begin{pmatrix} \lambda_{t+1}^h \\ \lambda_{t+1}^f \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \cdot \begin{pmatrix} \lambda_t^h \\ \lambda_t^f \end{pmatrix} + \begin{pmatrix} \epsilon_t^h \\ \epsilon_t^f \end{pmatrix}$$

Market Clear

$$y_t^i = c_t^i + x_t^i + (z_{t+1}^i - z_t^i) + \underline{nx_t^i}$$ 净出口

$$y_t^h + y_t^f = c_t^h + c_t^f + x_t^h + x_t^f + (z_{t+1}^h - z_t^h) + (z_{t+1}^f - z_t^f)$$ 全球出清

Solution:

Social Planner:

$$\text{maximize}: \quad E_0 \cdot \sum_{t=0}^{\infty} \beta^t \cdot \left[w_h \cdot U(c_t^h, h_t^h) + w_f \cdot U(c_t^f, h_t^f) \right]$$

$$L = E_0 \cdot \sum_{t=0}^{\infty} \beta^t \cdot \Bigg[\sum_i w_i \cdot U(c_t^i, h_t^i)$$

$$+ \Lambda_t \cdot \left(\sum_i y_t^i - \sum_i c_t^i - \sum_i z_{t+1}^i + \sum_i z_t^i - \sum_i \sum_j^J \phi_j \cdot s_{jt}^i \right)$$

$$+ \sum_i Q_t^i \cdot \left[(1-\delta) k_t^i + s_{1t}^i - k_{t+1}^i \right]$$

$$+ \sum_i \sum_j^{J-1} \pi_{jt}^i \cdot \left(s_{j+1,t}^i - s_{j,t+1}^i \right)$$

$$+ \sum_i \cdot P_t^i \cdot \left[(1-\eta) a_t^i + n_t^i - a_{t+1}^i \right] \Bigg]$$

Consumption FOC.

$$\frac{\partial L}{\partial C_t^i} = w_i \cdot U_c(C_t^i, h_t^i) - \Lambda_t = 0$$

$$\Lambda_t = w_i \cdot U_c(C_t^i, h_t^i) \quad ①$$

Labor FOC

$$\frac{\partial L}{\partial n_t^i} = w_i \cdot U_h(C_t^i, h_t^i) \cdot \frac{\partial h_t^i}{\partial n_t^i} + \Lambda_t \cdot MPL_t^i + P_t^i = 0$$

$$\begin{cases} h_t^i = 1 - \alpha n_t^i - (1-\alpha) \eta \cdot a_t^i \\ \frac{\partial h_t^i}{\partial n_t^i} = -\alpha \end{cases}$$

$$\alpha \cdot w_i \cdot U_h(C_t^i, h_t^i) = \Lambda_t \cdot MPL_t^i + P_t^i \quad ②$$

a_{t+1} FOC

$$\frac{\partial L}{\partial a_{t+1}^i} = \beta^t \cdot (-P_t^i) + \beta^{t+1} \cdot \left[W_i \cdot U_h(C_{t+1}^i \cdot h_{t+1}^i) \cdot \frac{\partial h_{t+1}^i}{\partial a_{t+1}^i} + P_{t+1}^i \cdot (1-\eta) \right] = 0$$

$$P_t^i = \beta \cdot \left[(1-\eta) P_{t+1}^i - (1-\alpha)\eta \cdot W_i \cdot U_h(C_{t+1}^i \cdot h_{t+1}^i) \right] \quad ③$$

Inventory FOC

$$\frac{\partial L}{\partial Z_{t+1}^i} = \beta^t \cdot \Lambda_t \cdot (-1) + \beta^{t+1} \cdot \Lambda_{t+1} \cdot \left(MPZ_{t+1}^i + 1 \right) = 0$$

$$\Lambda_t = \beta \cdot \Lambda_{t+1} \cdot \left(MPZ_{t+1}^i + 1 \right) \quad ④$$

Capital FOC

$$\frac{\partial L}{\partial k_{t+1}^i} = \beta^t \cdot (-Q_t^i) + \beta^{t+1} \cdot \left[\Lambda_{t+1} \cdot MPK_{t+1}^i + Q_{t+1}^i \cdot (1-\delta) \right] = 0$$

$$Q_t^i = \beta \cdot \left[\Lambda_{t+1} \cdot MPK_{t+1}^i + (1-\delta) Q_{t+1}^i \right] \quad ⑤$$

S_{1t} FOC（即将完工项目会进入资本项目，因此有 Q_t）

$$\frac{\partial L}{\partial S_{1t}} = \beta^{t-1} \cdot \pi_{1,t-1}^{i} (-1) + \beta^{t} \cdot \left[\Lambda_t \cdot (-\phi_1) + Q_t^{i} \right] = 0$$

$$\pi_{1,t-1}^{i} = \beta \cdot \left(Q_t^{i} - \phi_1 \Lambda_t \right) \quad 6^1$$

S_{jt}, $j = 2, \cdots J-1$（正常滚动项目）

$$\frac{\partial L}{\partial S_{jt}} = \beta^{t-1} \cdot \left(-\pi_{j,t-1}^{i} \right) + \beta^{t} \cdot \left(\Lambda_t \cdot (-\phi_j) + \pi_{j+1,t}^{i} \right) = 0$$

$$\pi_{j,t-1}^{i} = \beta \cdot \left(\pi_{j+1,t}^{i} - \Lambda_t \phi_j \right) \quad 6^2$$

S_{Jt} FOC（新开工项目不跨期）

$$\frac{\partial L}{\partial S_{Jt}} = \Lambda_t (-\phi_J) + \pi_{J+1,t}^{i} = 0$$

$$\Lambda_t \cdot \phi_J = \pi_{J+1,t}^{i} \quad 6^3$$

消去 π：

$$\begin{cases} \pi_{J-1,t} = \Lambda_t \cdot \phi_J & ⑥^1 \\ \pi_{j,t} = \beta \cdot \left(\pi_{j-1,t+1} - \Lambda_{t+1} \phi_j\right) & j = 2, \dots J-1 \quad ⑥^2 \\ \pi_{1,t} = \beta \cdot \left(Q_{t+1} - \Lambda_{t+1} \cdot \phi_1\right) & ⑥^3 \end{cases}$$

由②式，令 $j = J-1$

$$\pi_{J-1,t} = \beta\left(\pi_{J-2,t+1} - \Lambda_{t+1} \phi_{J-1}\right)$$

代入①式

$$\Lambda_t \cdot \phi_J = \beta \cdot \left(\pi_{J-2,t+1} - \Lambda_{t+1} \cdot \phi_{J-1}\right)$$

$$\Rightarrow \pi_{J-2,t+1} = \beta^0 \cdot \Lambda_{t+1} \cdot \phi_{J-1} + \beta^{-1} \cdot \Lambda_t \phi_J \quad ④$$

由②式，令 $j = J-2$

$$\pi_{J-2,t} = \beta\left(\pi_{J-3,t+1} - \Lambda_{t+1} \phi_{J-2}\right)$$

$$\pi_{J-2,t+1} = \beta\left(\pi_{J-3,t+2} - \Lambda_{t+2} \phi_{J-2}\right)$$

代入④

$$\beta^0 \cdot \Lambda_{t+1} \cdot \phi_{J-1} + \beta^{-1} \cdot \Lambda_t \phi_J = \beta\left(\pi_{J-3,t+2} - \Lambda_{t+2} \phi_{J-2}\right)$$

$$\Rightarrow \pi_{J-3,t+2} = \beta^0 \cdot \Lambda_{t+2} \cdot \phi_{J-2} + \beta^{-1} \cdot \Lambda_{t+1} \phi_{J-1} + \beta^{-2} \cdot \Lambda_t \cdot \phi_J$$

$$\vdots$$

$$\pi_{J-m-1,\, t+m} = \beta^0 \cdot \Lambda_{t+m} \cdot \phi_{J-m} + \beta^1 \cdot \Lambda_{t+m-1} \cdot \phi_{J-m+1} + \beta^2 \cdot \Lambda_{t+m-2} \cdot \phi_{J-m+2} + \cdots + \beta^{-m} \cdot \Lambda_t \, \phi_J$$

$$\pi_{J-m-1,\, t+m} = \sum_{r=0}^{m} \beta^{r-m} \cdot \Lambda_{t+r} \cdot \phi_{J-r}$$

令 $m = J-2$

$$\pi_{1,\, t+J-2} = \sum_{r=0}^{J-2} \beta^{r-J+2} \cdot \Lambda_{t+r} \cdot \phi_{J-r}$$

由 ③ 式

$$\pi_{1,\, t+J-2} = \beta \cdot \left(Q_{t+J-1} - \Lambda_{t+J-1} \phi_1 \right)$$

$$\Rightarrow \sum_{r=0}^{J-2} \beta^{r-J+2} \cdot \Lambda_{t+r} \cdot \phi_{J-r} = \beta \cdot \left(Q_{t+J-1} - \Lambda_{t+J-1} \phi_1 \right)$$

$$Q_{t+J-1} = \underbrace{\Lambda_{t+J-1} \cdot \phi_1}_{} + \sum_{r=0}^{J-2} \beta^{r-J+1} \Lambda_{t+r} \, \phi_{J-r}$$

$$\beta^0 \cdot \Lambda_{t+J-1} \cdot \phi_1 = \beta^{(J-1)-J+1} \Lambda_{t+(J-1)} \cdot \phi_{J-(J-1)}$$

$$Q_{t+J-1} = \sum_{r=0}^{J-1} \beta^{r-J+1} \cdot \Lambda_{t+r} \, \phi_{J-r}$$

$$\beta^{J-1} \cdot Q_{t+J-1} = \sum_{r=0}^{J-1} \beta^{r} \cdot \Lambda_{t+r} \, \phi_{J-r} \quad ⑥$$

消去 Q_t

$$\begin{cases} \beta^{J-1} \cdot Q_{t+J-1} = \sum_{r=0}^{J-1} \beta^{r} \cdot \Lambda_{t+r} \phi_{J-r} & ⑥ \\ Q_t^i = \beta \cdot \left[\Lambda_{t+1} \cdot MPK_{t+1}^i + (1-\delta) Q_{t+1}^i \right] & ⑤ \end{cases}$$

$$\beta^{-1} \cdot Q_t - (1-\delta) Q_{t+1} = \Lambda_{t+1} \cdot MPK_{t+1}$$

$$\beta^{-1} \cdot Q_{t+J-1} - (1-\delta) Q_{t+J} = \Lambda_{t+J} \cdot MPK_{t+J}$$

$$\beta^{-1} \cdot \beta^{J-1} \cdot Q_{t+J-1} - (1-\delta) \cdot \beta^{J-1} \cdot Q_{t+J} = \beta^{J-1} \cdot \Lambda_{t+J} \cdot MPK_{t+J}$$

$$\beta^{-1} \cdot \sum_{r=0}^{J-1} \beta^{r} \Lambda_{t+r} \phi_{J-r} - (1-\delta) \cdot \sum_{r=0}^{J-1} \beta^{r} \Lambda_{t+1+r} \cdot \phi_{J-r} = \beta^{J-1} \Lambda_{t+J} \cdot MPK_{t+J}$$

$$\sum_{r=0}^{J-1} \beta^{r} \Lambda_{t+r} \phi_{J-r} - (1-\delta) \sum_{r=0}^{J-1} \beta^{r+1} \Lambda_{t+1+r} \phi_{J-r} = \beta^{J} \Lambda_{t+J} MPK_{t+J}$$

$$\sum_{r=0}^{J-1} \beta^{r} \Lambda_{t+r} \phi_{J-r} - (1-\delta) \sum_{m=1}^{J} \beta^{m} \Lambda_{t+m} \phi_{J-r} = \beta^{J} \Lambda_{t+J} MPK_{t+J} \quad ⑦$$

消去 P_t :

$$\begin{cases} \beta(1-\delta)\cdot P_{t+1}^{i} - P_t^{i} = \beta(1-\alpha)\delta\, w_i \cdot U_n(c_{t+1}^{i}, h_{t+1}^{i}) & ③ \\ P_t^{i} = \alpha\, w_i \cdot U_n(c_t^{i}, h_t^{i}) - \Lambda_t \cdot MPL_t^{i} & ② \end{cases}$$

$$③\ LHS = \beta(1-\delta)\cdot\left[\alpha\cdot w_i\cdot U_n(c_{t+1}^{i}, h_{t+1}^{i}) - \Lambda_{t+1}\cdot MPL_{t+1}^{i}\right] - \left[\alpha\, w_i\cdot U_n(c_t^{i}, h_t^{i}) - \Lambda_t\cdot MPL_t^{i}\right]$$

$$= \alpha\beta(1-\delta) w_i\cdot U_n(t+1) - \beta(1-\delta)\Lambda_{t+1}\cdot MPL_{t+1}^{i} - \alpha w_i\cdot U_n(t) + \Lambda_t\cdot MPL_t^{i}$$

$$= \alpha\beta(1-\delta) w_i\cdot U_n(t+1) - \alpha w_i\, U_n(t) - \beta(1-\delta)\Lambda_{t+1}\cdot MPL_{t+1} + \Lambda_t\, MPL_t^{i}$$

$$③\ RHS = \beta(1-\alpha)\delta\, w_i\cdot U_n(t+1)$$

$$\Downarrow$$

$$(\alpha-\delta)\cdot\beta\cdot w_i\cdot U_n(t+1) - \alpha w_i\cdot U_n(t) = \beta(1-\delta)\Lambda_{t+1}\cdot MPL_{t+1} - \Lambda_t\, MPL_t^{i} \quad ⑧$$